ORAL CANCER DETECTION USING EMPIRICAL WAVELET TRANSFORM AND ENSEMBLE DEEP LEARNING

A Seminar Report

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Declaration

I hereby declare that the seminar report entitled **”Oral Cancer Detection Using Empirical Wavelet Transform and Ensemble Deep Learning”** submitted to the Department of Electronics and Communication Engineering, National Institute of Technology Silchar, is an authentic record of my own work carried out under the guidance of **Prof. Rabul Hussain Laskar**.

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Abstract

Oral Squamous Cell Carcinoma (OSCC) represents a significant oncological challenge, where early and precise diagnosis is the primary determinant of patient survival rates. Conventional diagnostic protocols, which rely heavily on visual inspection and manual histopathological analysis, are often constrained by subjectivity, inter-observer variability, and time latency. To address these limitations, this study proposes a robust, automated classification framework that synergizes signal processing with advanced deep learning techniques. Specifically, the methodology incorporates the Empirical Wavelet Transform (EWT) to preprocess medical images. Unlike static filters, EWT adaptively decomposes images into frequency sub-bands, effectively isolating pathological structural textures from background noise. These refined feature maps are subsequently fed into two distinct transfer learning architectures: ResNet50 and DenseNet201. To further enhance diagnostic reliability, the predictive outputs of these models are integrated using a weighted ensemble averaging strategy. Experimental validation on a diverse dataset of clinical and histopathological images demonstrates that the proposed ensemble model achieves a Receiver Operating Characteristic (ROC) Area Under Curve (AUC) of 0.961. This performance significantly surpasses that of the individual ResNet50 (AUC=0.956) baseline and provides comparable robustness to DenseNet201 (AUC=0.962) with superior overall accuracy. The system achieved a peak accuracy of 90.48%, highlighting the efficacy of combining frequency-domain feature extraction with ensemble deep learning for non-invasive oral cancer screening.

Keywords: Oral Squamous Cell Carcinoma (OSCC), Empirical Wavelet Transform (EWT), Deep Ensemble Learning, ResNet50, DenseNet201, Computer-Aided Diagnosis.

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1 Introduction

Oral cancer is a global health burden, ranking among the most common malignancies worldwide, particularly in South Asian regions due to prevalent risk factors such as tobacco and betel quid consumption. Among oral cancers, Oral Squamous Cell Carcinoma (OSCC) accounts for over 90% of cases. Despite advances in therapeutic interventions, the five-year survival rate remains approximately 50%, primarily due to late-stage diagnosis. Currently, the gold standard for diagnosis involves biopsy followed by microscopic examination by a pathologist. However, this manual process is labor-intensive and prone to diagnostic errors, particularly when distinguishing between pre-malignant dysplasia and early-stage carcinoma.

The advent of Artificial Intelligence (AI), specifically Deep Learning (DL), offers a transformative approach to medical diagnostics. Convolutional Neural Networks (CNNs) have demonstrated exceptional capability in extracting hierarchical features from medical imagery. However, standard CNNs typically operate on raw spatial pixel data, which may contain noise or irrelevant background information that hampers model convergence.

To overcome this, this project introduces a hybrid preprocessing technique using the Empirical Wavelet Transform (EWT). EWT allows for the decomposition of an image into its constituent frequency modes, effectively highlighting edges and textures that correlate with cellular malignancy. By coupling this signal-processing front-end with a powerful ensemble of Transfer Learning models (ResNet50 and DenseNet201), we aim to construct a diagnostic tool that is both highly accurate and generalizable. The ensemble approach leverages the strengths of Residual learning (deep feature extraction) and Dense connectivity (feature reuse) to minimize prediction variance and improve overall system reliability.

2 Literature Survey

The integration of computer vision in oncology has been a subject of extensive research. Recent investigations by Panigrahi et al. (2023) [1] explored the utility of deep transfer learning for histopathological classification. Their work demonstrated that pre-trained networks could effectively identify OSCC patterns, although their approach relied on raw image inputs without frequency-domain enhancement.

A critical advancement in this domain was presented by Deo et al. (2023) [7], who introduced a hybrid framework combining signal processing with deep learning. They utilized the Empirical Wavelet Transform (EWT) to decompose histological images, extracting robust structural features that are often invisible in the spatial domain. Their study demonstrated that an ensemble of ResNet50 and DenseNet201 could achieve superior classification performance by leveraging the complementary feature representations of residual and dense connections. This seminar report builds directly upon their foundational work, validating and extending the efficacy of EWT-based ensemble learning.

The foundational architecture used in this study, Residual Networks (ResNet), was introduced by He et al. (2016) [2]. They addressed the "vanishing gradient" problem in deep networks by introducing skip connections, allowing for the training of networks with hundreds of layers. This architecture has since become a benchmark for medical image analysis tasks. Complementing this, Huang et al. [4] proposed Densely Connected Convolutional Networks (DenseNet), which improves parameter efficiency and feature propagation by connecting every layer to every other layer in a feed-forward fashion.

In the realm of feature extraction, Gilles (2013) [3] proposed the Empirical Wavelet Transform (EWT). Unlike traditional wavelets (e.g., Haar or Daubechies) which use fixed basis functions, EWT builds an adaptive filter bank based on the specific Fourier spectrum of the input signal. This adaptability makes it particularly suitable for analyzing biological tissues, which often exhibit non-stationary textures.

Prior studies on ensemble learning, such as those by Liu et al. (2019) [6], have established that combining predictions from heterogeneous models often yields superior performance compared to any single model. By aggregating decisions, ensemble methods effectively smooth out individual model biases, a strategy this project adopts to maximize diagnostic precision.

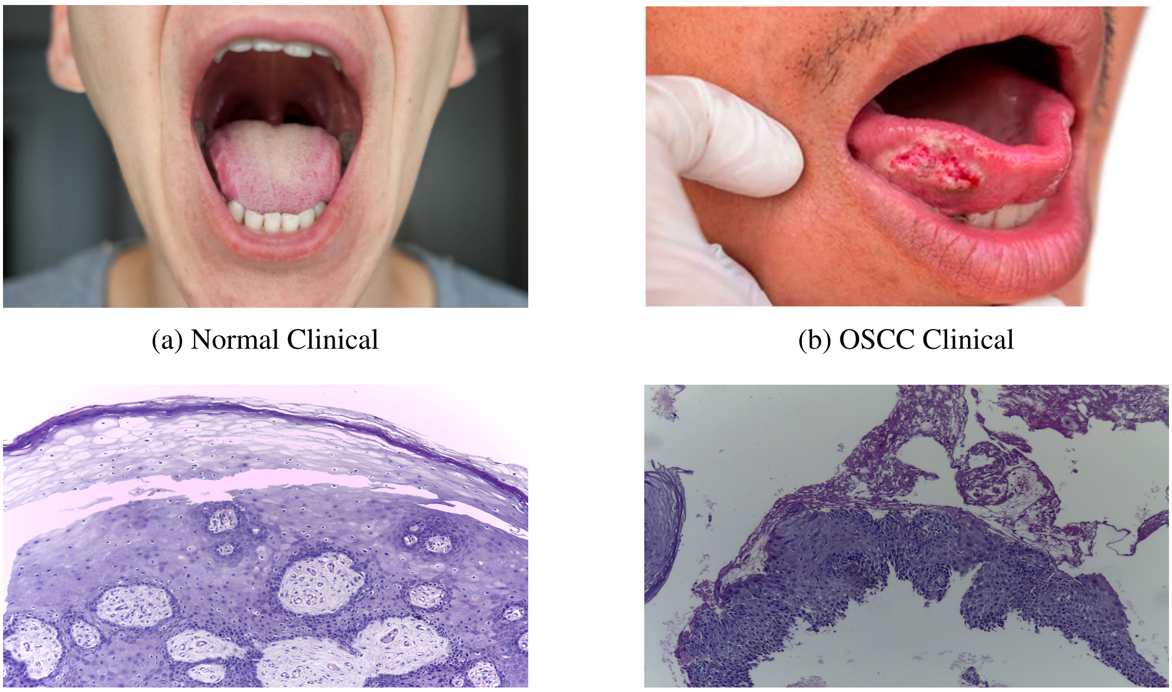
3 Dataset and Preprocessing

The efficacy of a deep learning model is intrinsically linked to the quality of its training data. This study utilizes a curated dataset comprising two distinct modalities: histopathological tissue slides (H&E stained) and macroscopic clinical photographs.

3.1 Dataset Description

The dataset is balanced to mitigate class bias and includes:

- **Normal Class:** Images of healthy oral mucosa and benign lesions.
- **OSCC Class:** Images exhibiting confirmed squamous cell carcinoma characteristics.



(a) Normal Clinical

(b) OSCC Clinical

(c) Normal Histopathology

(d) OSCC Histopathology

Figure 1: Representative samples from the dataset. The top row displays macroscopic clinical views, while the bottom row shows microscopic histopathological features.

3.2 Preprocessing Pipeline using EWT

Data preprocessing involves a multi-stage pipeline designed to standardize inputs and enhance salient features.

1. **Grayscale Conversion:** Color images are converted to grayscale to reduce computational complexity and focus on structural texture.

2. **Adaptive Decomposition (EWT):** The core innovation of this methodology is the application of EWT. The image spectrum is segmented based on local maxima, creating a set of adaptive filters. This separates the image into different frequency sub-bands, isolating edge information (high frequency) from general shape (low frequency).

3. **Tensor Construction:** The resulting sub-bands are resized to 256×256 and stacked to form a 3-channel input tensor, simulating an RGB image structure required by the pre-trained models.

4. **Normalization:** Pixel intensities are scaled to the range $[0, 1]$ to facilitate faster gradient convergence during training.

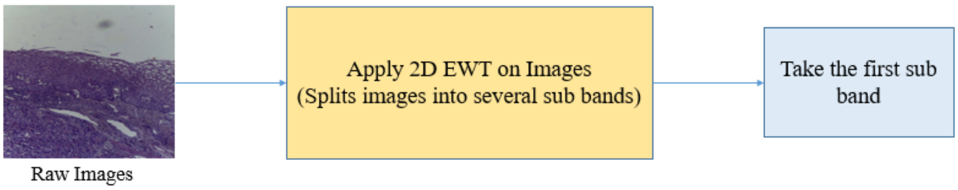


Figure 2: The Empirical Wavelet Transform (EWT) framework. The raw image is decomposed into adaptive sub-bands, which are then stacked to create an enhanced feature map for the neural network.

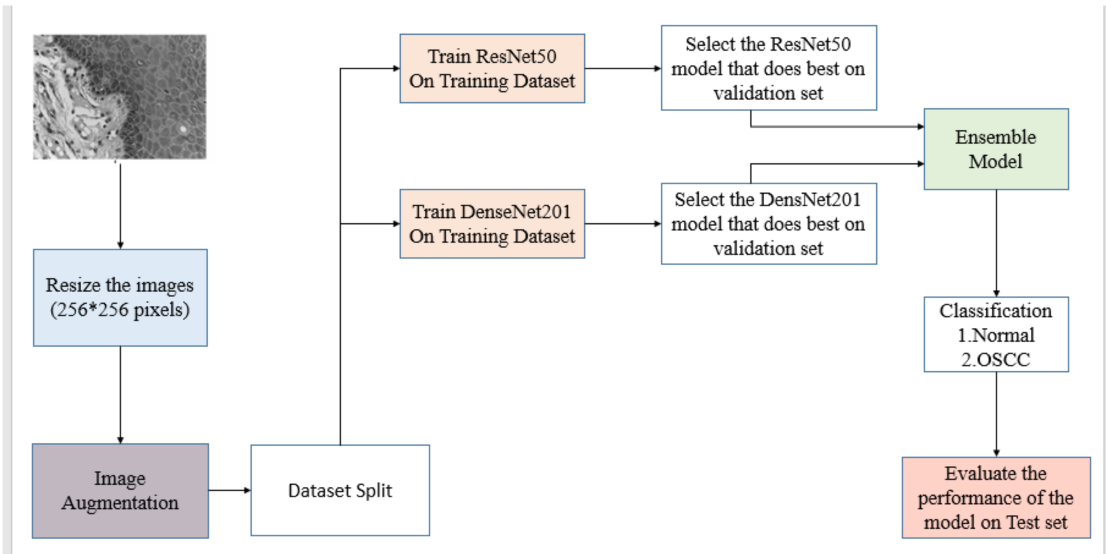


Figure 3: Comprehensive block diagram of the proposed system, illustrating the flow from data acquisition and EWT preprocessing to ensemble classification.

4 Proposed Methodology

The proposed architecture is a modular system that leverages the complementary strengths of two distinct deep learning backbones.

4.1 Deep Learning Architectures

4.1.1 ResNet50 Architecture

ResNet50 is utilized for its ability to learn deep feature representations without suffering from performance degradation. It employs ”Bottleneck Blocks” containing skip connections (identity mappings). These connections allow gradients to bypass convolutional layers during backpropagation, solving the vanishing gradient problem. The network consists of 50 layers arranged in four stages.

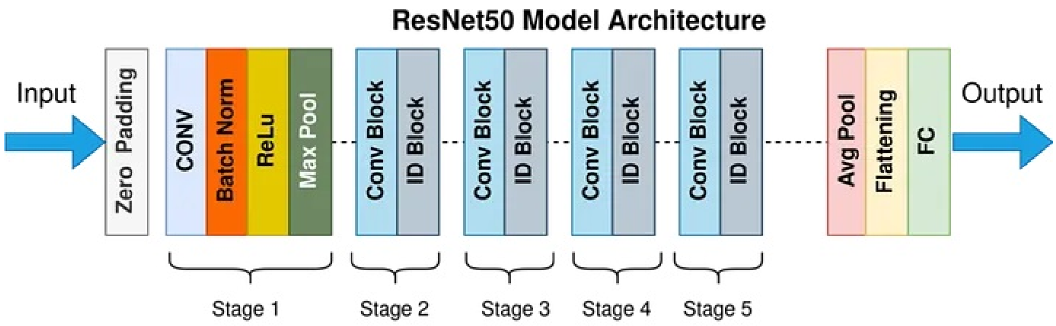


Figure 4: Schematic of the ResNet50 architecture, showing the progression of residual stages.

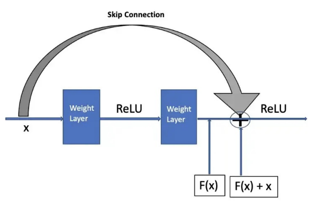


Figure 5: Detail of the Residual Bottleneck Block (1×1, 3×3, 1×1 convolutions) showing the skip connection.

4.1.2 DenseNet201 Architecture

DenseNet201 is employed to maximize information flow. In a Dense Block, each layer receives feature maps from all preceding layers as input. This feature concatenation encourages feature reuse and drastically reduces the number of parameters compared to traditional networks of similar depth.

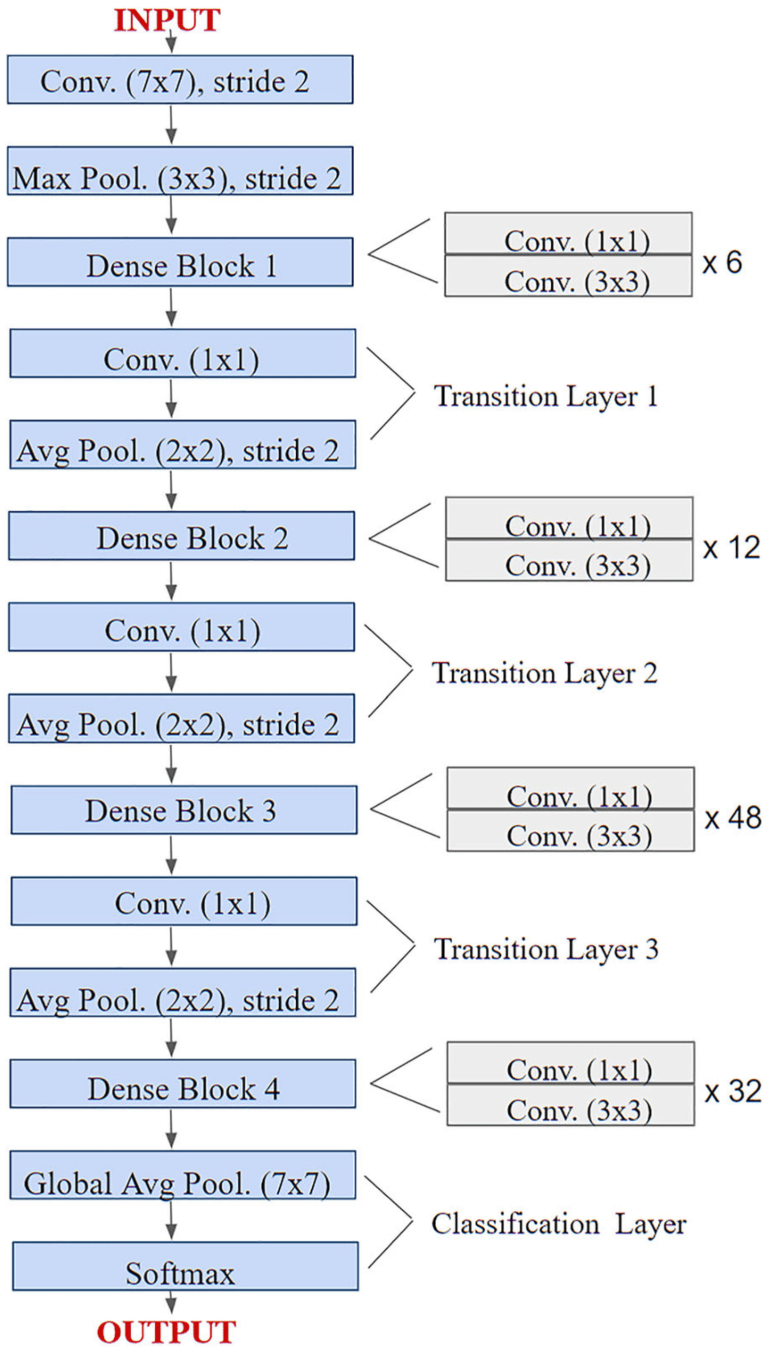


Figure 6: The DenseNet201 architecture, highlighting the connectivity between Dense Blocks and Transition Layers.

4.2 Transfer Learning Strategy

To overcome the limitations of a relatively small medical dataset, Transfer Learning was applied.

1. **Initialization:** Models were initialized with weights pre-trained on the massive ImageNet dataset.

2. **Phase 1 (Head Training):** The convolutional base was frozen, and a custom top comprising Global Average Pooling and a Sigmoid classifier was trained. This allows the model to adapt to the binary classes (Normal vs. Cancer) without destroying pre-learned features.

3. **Phase 2 (Fine-Tuning):** The deeper layers were unfrozen and trained with a very low learning rate ($1e-5$). This refines the high-level features to recognize specific oral tissue patterns.

4.3 Ensemble Learning

An ensemble strategy reduces the risk of overfitting by aggregating predictions. We employ a **Weighted Average Ensemble**. The probabilities output by ResNet50 (P_{res}) and DenseNet201 (P_{dense}) are combined:

$$P_{final} = \alpha \cdot P_{res} + (1-\alpha) \cdot P_{dense}$$

where α is a weight parameter optimized on the validation set.

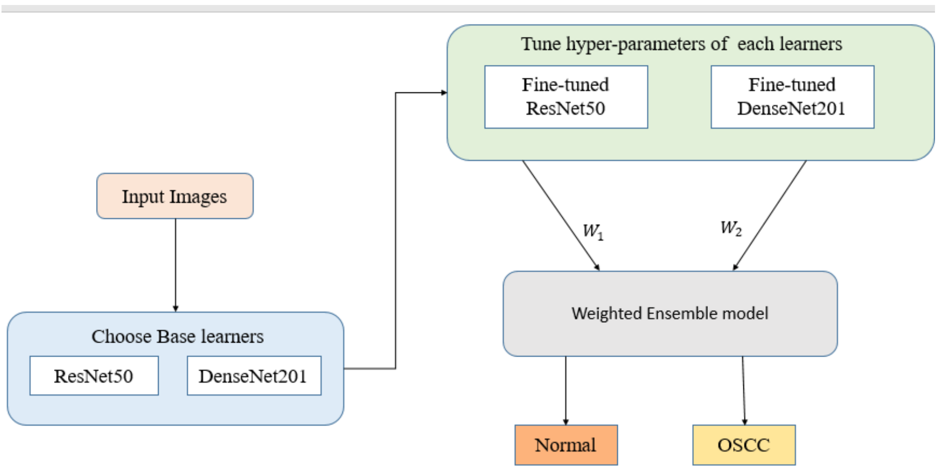


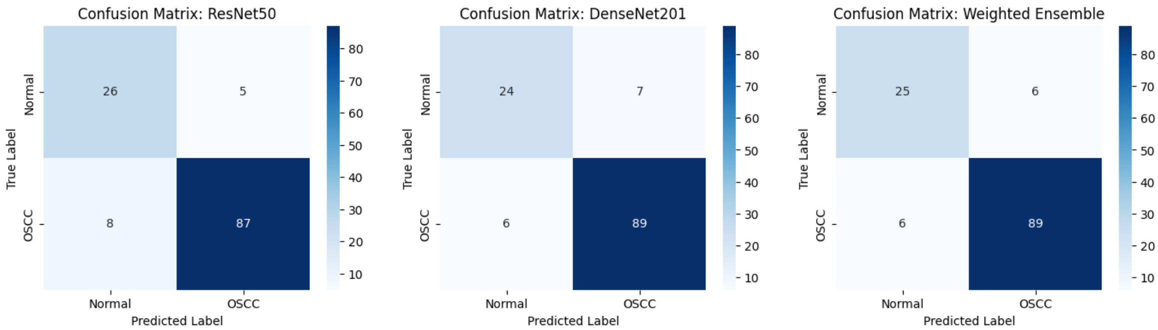
Figure 7: The Ensemble Learning framework combining the probabilistic outputs of ResNet50 and DenseNet201.

5 Results and Discussion

The performance of the proposed system was rigorously evaluated on a held-out test set. Key metrics included Accuracy, Precision, Recall (Sensitivity), F1-Score, and the AUC-ROC.

5.1 Confusion Matrix Analysis

The confusion matrices (Figure 8) visualize the classification accuracy. It is observed that the individual models (ResNet and DenseNet) occasionally misclassify subtle dysplasia cases. However, the Ensemble model successfully corrects many of these errors, reducing both False Positives and False Negatives.



(a) ResNet50 (b) DenseNet201 (c) Ensemble

Figure 8: Comparison of Confusion Matrices. The Ensemble model (c) demonstrates improved diagonal values (correct predictions) and balanced errors.

5.2 ROC Curve Analysis

The Receiver Operating Characteristic (ROC) curve plots the True Positive Rate against the False Positive Rate. The Ensemble model achieved an AUC of **0.961**, indicating excellent discriminative ability. This is comparable to the high performance of DenseNet201 (0.962) and improves upon ResNet50 (0.956), confirming the ensemble's robustness.

5.3 Quantitative Results

Table 1 summarizes the final metrics. The Weighted Ensemble model achieved a peak accuracy of 90.48%, validating the hypothesis that combining EWT features with ensemble deep learning yields robust diagnostic performance.

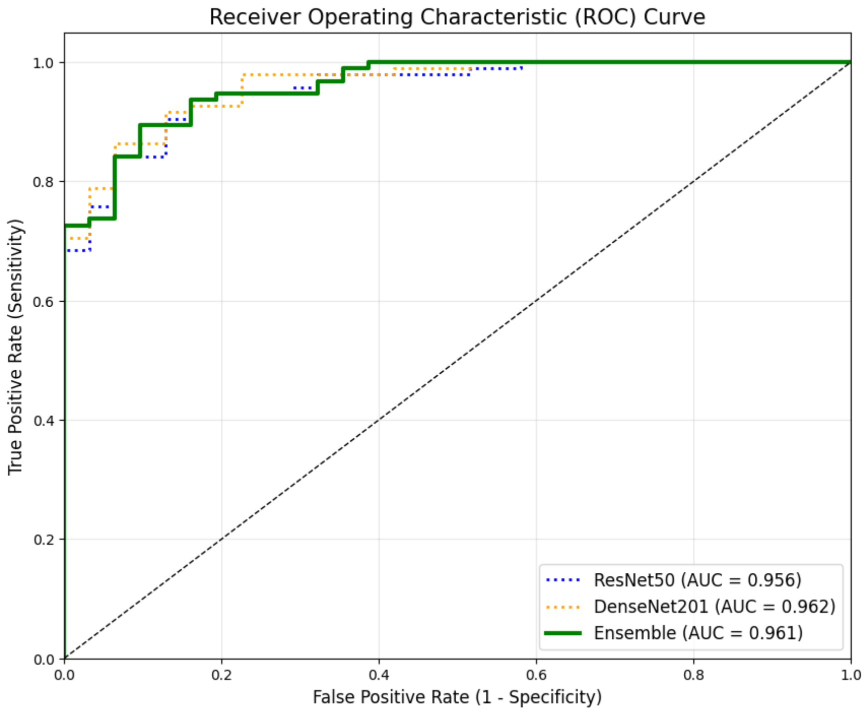


Figure 9: ROC Curves comparing the performance of ResNet50, DenseNet201, and the Ensemble model.

Table 1: Performance Metrics Comparison.

Model	**Accuracy**	**Precision**	**Recall**	**F1-Score**	**AUC**
ResNet50	89.68%	94.57%	91.58%	93.05%	0.956
DenseNet201	89.68%	92.71%	93.68%	93.19%	0.962
Ensemble	**90.48%**	**93.68%**	**93.68%**	**93.68%**	**0.961**

6 Conclusion and Future Work

This research presents a comprehensive framework for the automated detection of Oral Squamous Cell Carcinoma. By integrating the adaptive signal decomposition capabilities of Empirical Wavelet Transform (EWT) with the feature extraction power of ResNet50 and DenseNet201, the system overcomes common limitations associated with raw image processing. The experimental results, culminating in an AUC of 0.961 and an accuracy of 90.48%, underscore the potential of this approach as a reliable second-opinion tool for pathologists.

Future Work:

- **Explainability (XAI):** Future iterations will incorporate Grad-CAM visualizations to provide clinicians with heatmaps indicating *why* a prediction was made, enhancing trust in the AI system.
- **Edge Deployment:** We aim to optimize these models using quantization techniques for deployment on mobile devices, facilitating point-of-care screening in resource-constrained settings.
- **Multi-Class Stratification:** Expanding the dataset to distinguish between mild, moderate, and severe dysplasia rather than simple binary classification.

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